Fuzzy Speed Control of a DC Motor with Saturation, Dead Zone and Friction

Fuzzy Control, Project 2

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Plant. Armature current i and speed ω follow

$$L\dot{i} = f_{DZ}(V) - Ri - K_e(i)\,\omega, \qquad J\dot{\omega} = K_t(i)\,i - B\omega - \tau_c\,\mathrm{sign}(\omega) - T_L,$$

with magnetic saturation $K_{e,t}(i) = K_{e0,t0}\,i/\sqrt{i^2 + I_s^2}$, an input dead zone of $V_{DZ} = 2\,\mathrm{V}$ and $|V| \le 24\,\mathrm{V}$. Values: $R = 1\,\Omega$, $L = 0.5\,\mathrm{H}$, $K_{e0} = K_{t0} = 0.05$, $J = 0.01\,\mathrm{kg\,m^2}$, $B = 0.003\,\mathrm{N\,m\,s}$, $I_s = 5\,\mathrm{A}$, Coulomb friction $\tau_c = 0.2\,\mathrm{N\,m}$ and load $T_L = 0.2\,\mathrm{N\,m}$ (0.3 in the SMC case). The task is to track $\omega^\star = 50\,\mathrm{rad/s}$. All simulations use explicit Euler with $\Delta t = 10^{-4}\,\mathrm{s}$.

Controllers (one script each):

- `Q_OpenLoop`: 24 V applied; the motor settles near 120 rad/s, which shows the need for feedback.
- `Q_PIDClassic`: PID with $K_p = 10$, $K_i = 2$, $K_d = 2$, as a baseline.
- `Q_Linear`: a fuzzy controller designed as for a linear plant. It has one input (the error), 31 triangular sets on $[-50, 50]$, and a centre-average output on $[-240, 240]\,\mathrm{V}$ before saturation, so it acts like a nonlinear proportional controller.
- `Q_FuzzyPID`: a fuzzy supervisor tunes $K_p \in [1, 10]$, $K_i \in [0, 5]$ and $K_d \in [0, 5]$ online from the error and its derivative (7 sets each, three 7×7 rule tables).
- `Q_SMC_Lambda`: sliding-mode control $V = K\,\mathrm{sat}(s/\phi)$ with $s = \lambda e + \dot{e}$, $K = 20$ and $\phi = 0.5$, where a 7-rule fuzzy system adapts the slope $\lambda \in [0.1, 10]$.

Results (re-run in MATLAB R2023b; `tools/step_metrics.m`):

Controller	Rise 10–90% (s)	Overshoot (%)	Settling 2% (s)	Final error (rad/s)
Classic PID	0.97	16.6	12.6	−0.22
Fuzzy, one input (10 s run)	0.97	2.7	not settled	3.31
Fuzzy-tuned PID	0.97	37.9	7.3	0.00
SMC with fuzzy λ	1.95	0	3.3	0.12

The single-input fuzzy controller has no integral action, so the load and friction leave a steady-state error. The fuzzy-tuned PID removes the error and settles faster than the fixed PID, but it overshoots more while the input is saturated at the start. Sliding-mode control with the fuzzy slope gives no overshoot and the shortest settling time; the boundary layer leaves a small error of 0.12 rad/s.

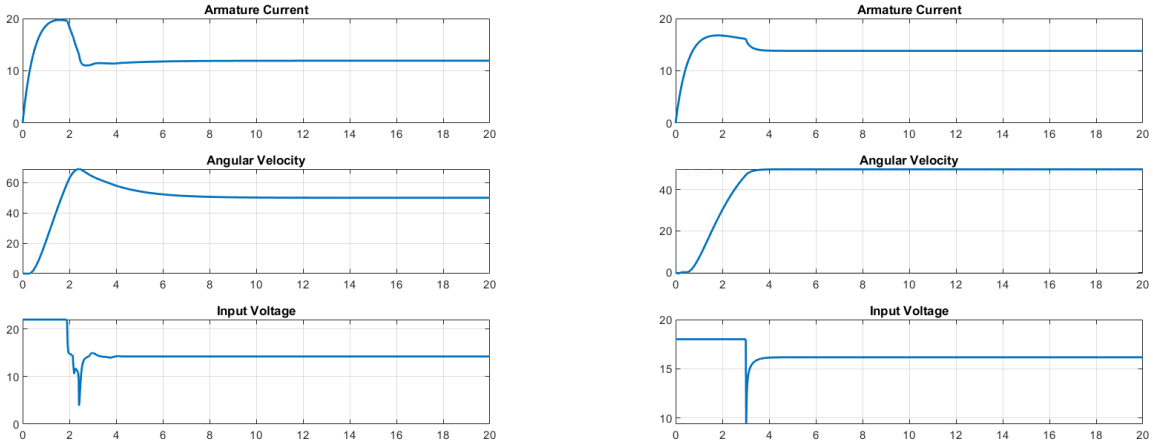


Figure 1: Current, speed and voltage with the fuzzy-tuned PID (left) and SMC with fuzzy λ (right).